

# VLSI Design for Manufacturability HW3

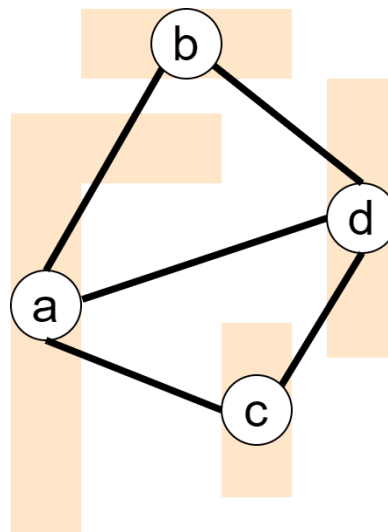
Deadline: 2025/5/1

## Instructions

- Ensure your handwriting is legible, or use a typed document if preferred.
- Staple all pages together before submission.

1. Refer to p. 1 to p.28, unit 7.

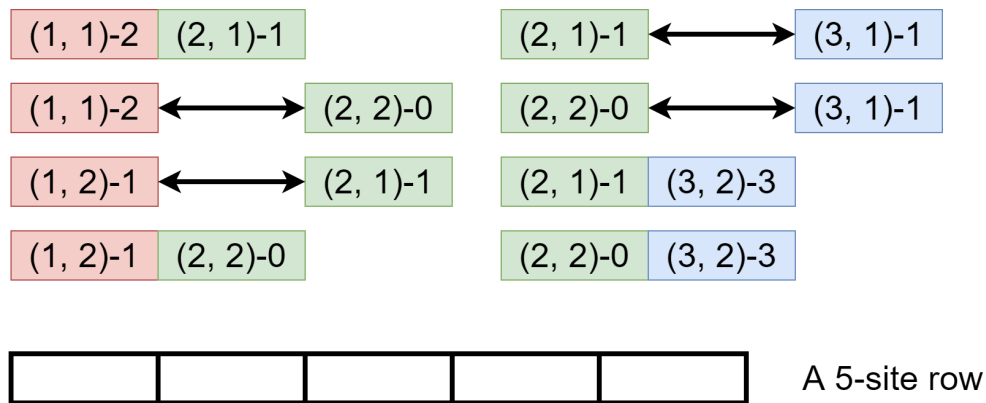
- a. The figure below shows a layout with four polygons (a, b, c, and d) and the corresponding conflict graph. Apply the polygon dummy extension to this layout. Draw the updated layout showing the extended areas of the polygons. (10 points)



- b. Continue problem 1.a, construct the solution graph for this layout. (15 points)
- c. In the TPL decomposition with stitches, what is the worst-case time complexity for finding a feasible decomposition for a row with  $N$  polygons, assuming there are at most  $L$  stitch candidate positions per polygon? (15 points)

2. Refer to unit 8.

- a. Three cells  $c_1$ ,  $c_2$ , and  $c_3$  are to be placed in a row with 5 sites and the order of cells is  $(c_1, c_2, c_3)$ . In the figure below, the cell with label  $(i, j)-k$  means  $c_i$  is assigned to color  $j$  with  $k$  stitches. The minimum spacing required between cells with each coloring combination is also shown in the figure below. Construct the graph for color assignment and cell placement (10 points), and solve it as a shortest path problem (5 points).



- b. Compare the time complexity of the 2-stage heuristic against the simultaneous approach. (15 points)
- c. Refer to p.20, explain why TPLPlacer-SPD achieves a lower stitch count compared to TPLPlacer. (15 points)
3. Refer to unit 9, In Table III of [1], the proposed approach achieves 9% fewer vias compared to [2]. In addition, [1] introduces no stitches or coloring conflicts, whereas [2] results in a significant number of stitches. Explain why [1] could achieve these improvements. (15 points)

[1] Double Patterning-Aware Detailed Routing with Mask Usage Balancing

[2] Triple Patterning Aware Routing and Its Comparison with Double Patterning Aware Routing in 14nm Technology